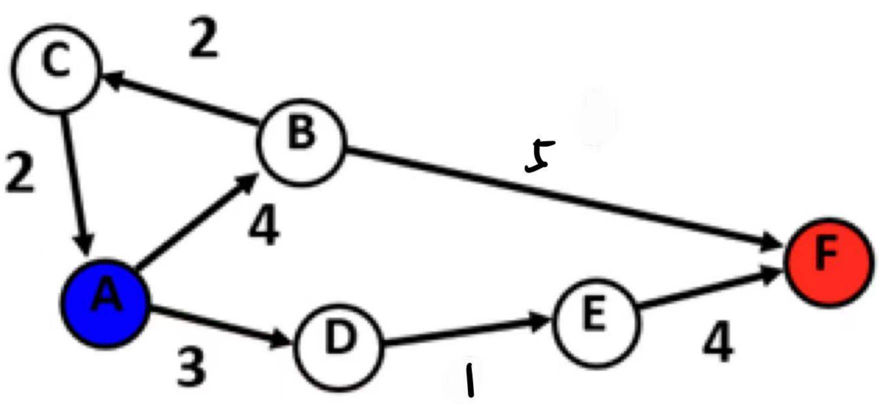
1.Search



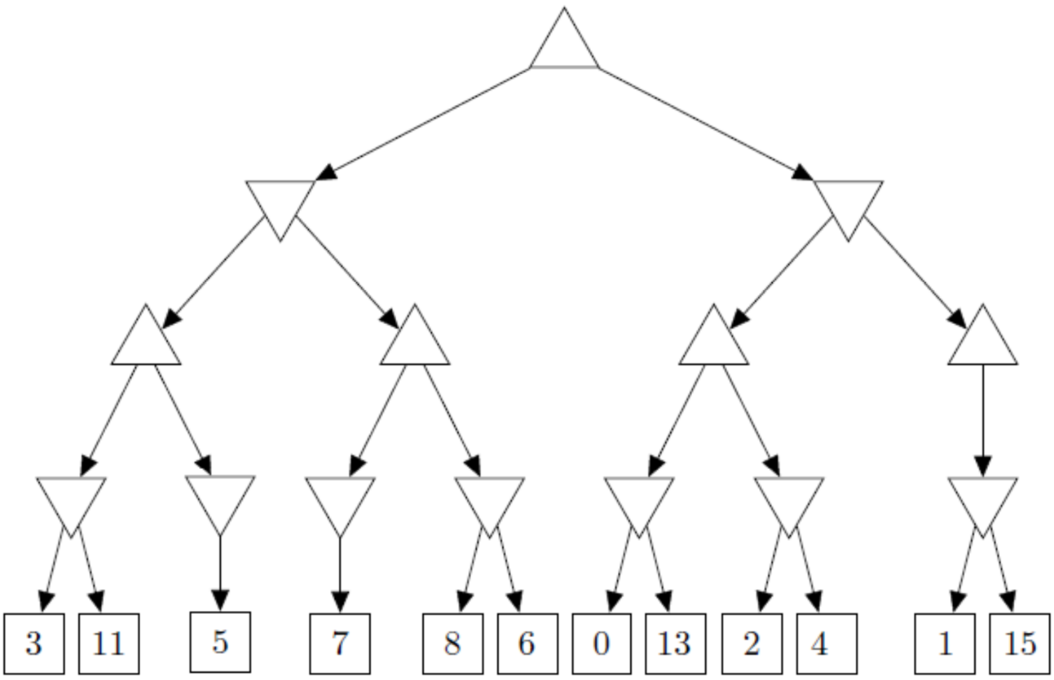
h(A) = 9   h(B) = 3   h(C) = 7 h(D) = 4 h(E) = 3   h(F) = 0

（1）Best first search

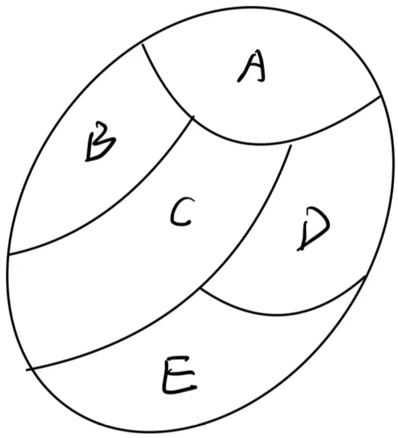
（2）A* search

（3）is h admissible

2.剪枝



3. CSP



着色问题 色彩空间：{R，B，Y}

(1) draw constraint graph

(2) A = R forward checking , the domain of B,C,D,E

(3) A = R arc consistency , the domain of B,C,D,E

4. reasoning

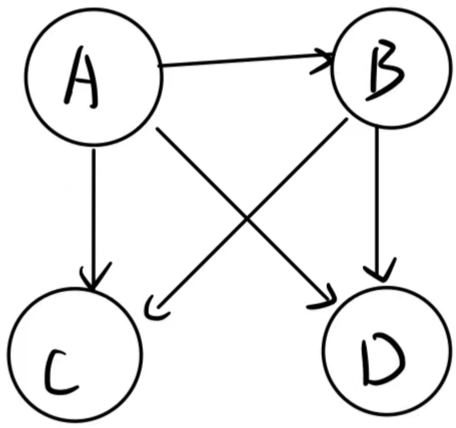
$$R => U \quad U => (\neg W) \quad \neg R => \neg W$$

Proof: $\neg W$

5. decision tree

A	B	C	Y(output)
F	F	F	F
T	F	T	T
T	T	F	T
T	T	T	F

6. Bayesian Network



概率（未特别说明指正确概率）

P(A) = 0.2

P(B|A):

A	P(B)
T	0.8
F	0.7

P(C|A,B) P(D|A,B)

A	B	P(C)	P(D)
T	T	0.4	0.9
T	F	0.2	0.8
F	T	0.3	0.1
F	F	0.1	0.2

(1) 变量消元求 $P(c|\neg d)$

(2) C ,D 条件独立性？ 给定A、B

7. CNN

（1） input feature map $30 \times 30 \times 6$ 池化：2×2 stride=2 问输出feature map尺寸

0	2	4	4
0	2	6	2
1	7	3	8
2	2	4	5

（2）上述map，（1）中池化层 maximum pool

（3）同上average poll

8. MDP

A	B	C
D	E	F（food）

living reward =0 ;blocked by outer wall;

A = {North,East,South,West}

noise = 0

discount = 0.5

(1) optimal policy

state	$\pi(state)$
A	
B	
C	
D	
E	
F	

(2) Value iteration

k	A	B	C	D	E	F
0	0	0	0	0	0	0
1						
2						
3						
4						

(3) optimal value of A

which k iteration $V_k(A) = V^*(A)$